Muhammad Ahmed

ML/AI Engineer | Software Engineer

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SUMMARY

ML/AI Engineer and **final-year Software Engineering student** at **FAST-NUCES**, specializing in **multi-agent LLM systems**, **RAG pipelines**, and **production fine-tuning (LoRA/QLoRA)**. Delivered **real-world agentic workflows** using **LangGraph** and **Redis** that cut **manual processing by 60–70%**. Experienced across the **full AI lifecycle** from **transformer fine-tuning** and **vector search** to **FastAPI deployment** and **HITL safety controls**.

SKILLS

Programming
Python, JavaScript, C++, C, Java

Agentic AI
LangChain, LangGraph, RAG, HITL, multi-agent orchestration, prompt engineering

Web Development
Next.js, React.js, Node.js, Tailwind CSS

ML/AI
TensorFlow, Scikit-learn, Unsloth, PEFT, LoRA/QLoRA, NLTK, Hugging Face Transformers

VectorDB & Data
ChromaDB, Supabase, PostgreSQL, MongoDB, Redis, Firebase

Deployment
FastAPI, Docker, REST APIs, Git, CI/CD basics

Work Experience

AI Product Developer Intern, *Techie Minions* 07/2026 – 08/2026
Lahore, Pakistan

- Built and shipped features for **Frameworq** , a **web-based project management platform**, adapting to evolving client requirements using **Claude Code** and **Claude Platform**
- Owned the full **features lifecycle** from changing client requirements to **AI-assisted development** to **production deployment** and verification
- Built **AI agents** to support core product functionality within **Frameworq** , **Automated testing** and workflow integration using **MCP servers** (**Playwright**, **Jira**, **Slack**) to verify releases in production

Generative AI Intern, *Arch Technologies* 01/2026 – 02/2026
Lahore, Pakistan

- **Fine-tuned domain-specific LLMs** using **LoRA/QLoRA**, reducing **hallucination rate** and improving **task accuracy** for **production chatbot deployments**.
- **Architected end-to-end RAG pipelines** from **document ingestion** and **chunking** to **vector retrieval** and **response generation** enabling **context-aware answers** over **proprietary knowledge bases**.
- Automated **LLM output evaluation** using **RAGAS metrics**, reducing **manual review time 60%** and improving **structured output precision by 18%**.

PROJECTS

AI Insurance Adjustor 02/2026 – 03/2026

- Engineered a **multi-agent claims pipeline** with **Computer Vision Models** , **RAG**, and LLMs to automate liability assessment, policy validation, and payout decisions cutting manual processing steps by **~70%**
- Designed an **async Redis message bus** coordinating **7 specialized agents** with persistent state and fault recovery, enabling **zero-downtime** claims processing

Multi Agent HR Assistant 01/2026 – 03/2026

- Built **multi-agent HR service desk** with **LangGraph orchestration** and **RAG-based policy retrieval** cutting response latency by **~65%** vs. manual lookup
- Built production **HITL safety layer** from scratch to override controls, **immutable audit logs**, and execution gates for compliant, accountable AI decisions

Self Improving Resoning Agent 02/2026 – 03/2026

- Built an **LLM critic system** to detect **reasoning errors** and **inconsistencies** in model outputs
- Enabled **autonomous self-correction** by integrating **critique mechanisms** into **evaluation pipelines**

EDUCATION

Software Engineering, *FAST (NUCES)* 08/2023 – Present
Pakistan

Certificates

TensorFlow Developer, Deep Learning Specialization, Machine Learning Specialization, IBM RAG & Agentic AI - Coursera (2025)

Awards

- 3rd place, UCP TAAKARA Web Hackathon - competed against 50+ teams. built and shipped a full-stack web application within a 8-hour window